Experiment – 7: Continuous-Time Fourier Transform and Spectrum

Objective 2:

To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

Experimental Procedure:

- Define t and the sampling interval.
- Generate x(t)=e^(−t^2)cos(2π3t).
- Compute the FFT-based transform with N = 4096.
- Plot the signal, magnitude and unwrapped phase.
- Identify the principal spectral peaks.

Code:

```
clc;
clear;
close all;

% Magnitude and Phase Spectrum of CT Signal
t = -2:0.001:2;
dt = t(2)-t(1);

% Input signal
x = exp(-t.^2).*cos(2*pi*3*t);

% FFT calculation
N = 4096;
X = fftshift(fft(x,N))*dt;
f = (-N/2:N/2-1)/(N*dt);
```

```
% Magnitude and Phase
magX = abs(X);
phaseX = unwrap(angle(X));

% Plotting
figure;

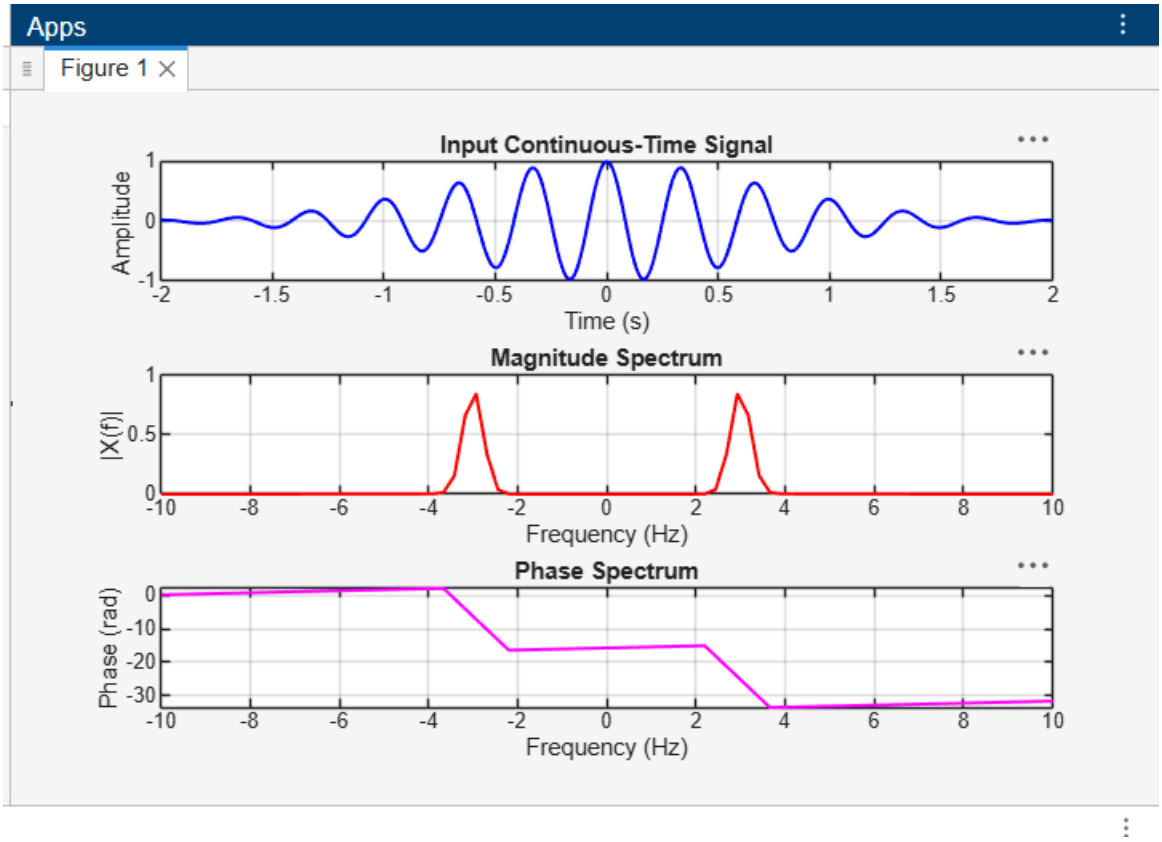
subplot(3,1,1);
plot(t,x,'b','LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Input Continuous-Time Signal');

subplot(3,1,2);
plot(f,magX,'r','LineWidth',1.2);
xlim([-10 10]);
grid on;
xlabel('Frequency (Hz)');
ylabel('|X(f)|');
title('Magnitude Spectrum');

subplot(3,1,3);
plot(f,phaseX,'m','LineWidth',1.2);
xlim([-10 10]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
```

title('Phase Spectrum');

output:



Objective 3:

Verification of Fourier Transform Properties and Frequency-Domain Characteristics

Experimental Procedure:

• Generate x(t)=e^(−t^2).

• Create x(t−0.4) using interpolation.

• Compute the FFT of the shifted signal.

• Form the theoretical transform using the time-shift property.

• Compare computed and theoretical magnitude and phase.

Code:

clc;

clear;

close all;

```
% Verification of Fourier Transform Properties
t = -2:0.001:2;
dt = t(2)-t(1);

% Input signal
x = exp(-t.^2);

% Time-shifted signal
t0 = 0.4;
x_shift = exp(-(t-t0).^2);

% FFT calculation
N = length(t);
f = (-N/2:N/2-1)/(N*dt);

X = fftshift(fft(x))*dt;
X_shift = fftshift(fft(x_shift))*dt;

% Theoretical time-shift property
X_theory = X.*exp(-1j*2*pi*f*t0);

% Plotting
figure;

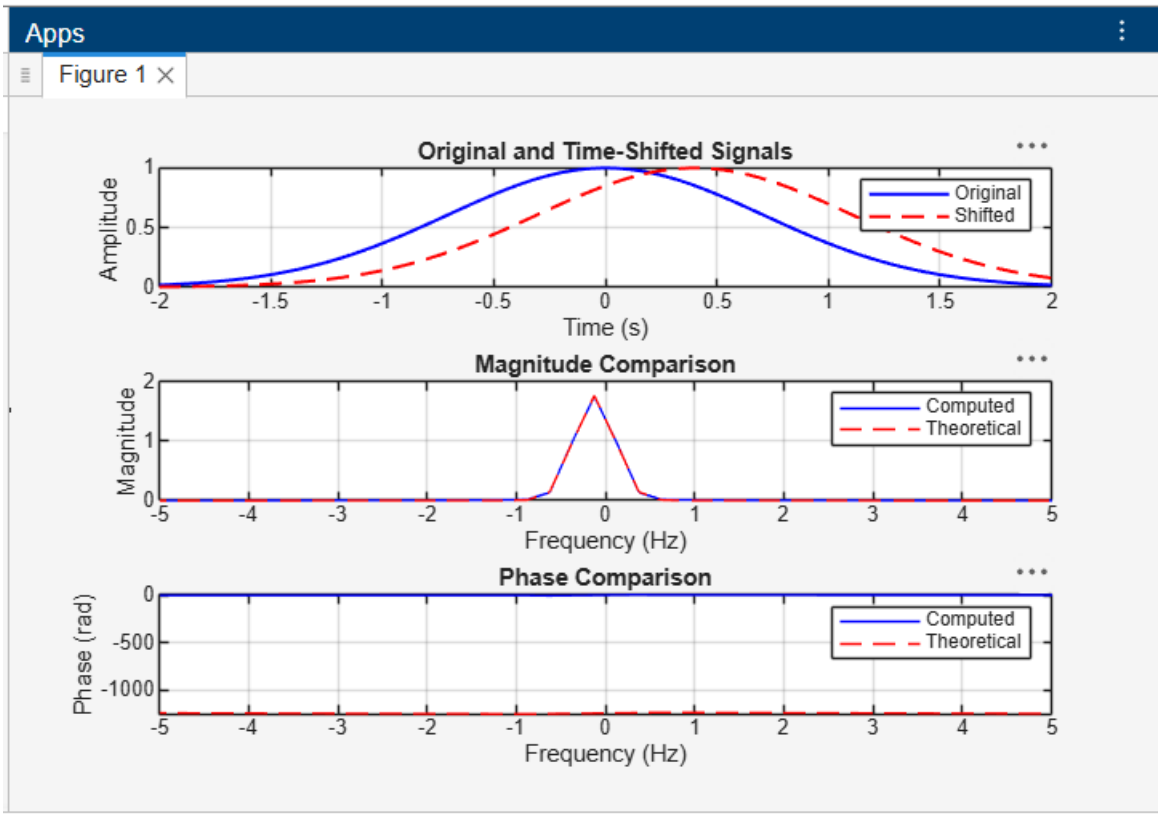
subplot(3,1,1);
plot(t,x,'b',t,x_shift,'r--','LineWidth',1.2);
grid on;
```

```
xlabel('Time (s)');
ylabel('Amplitude');
title('Original and Time-Shifted Signals');
legend('Original','Shifted');

subplot(3,1,2);
plot(f,abs(X_shift),'b',f,abs(X_theory),'r--');
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Magnitude Comparison');
legend('Computed','Theoretical');

subplot(3,1,3);
plot(f,unwrap(angle(X_shift)),'b', ...
    f,unwrap(angle(X_theory)),'r--');
xlim([-5 5]);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('Phase Comparison');
legend('Computed','Theoretical');
```

output:



Result:

Thus, the magnitude and phase spectra of continuous-time signals were successfully obtained using FFT, and the time-shifting property of the Fourier Transform was verified by comparing the computed and theoretical frequency-domain characteristics.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(ii) To determine the magnitude and phase spectra of continuous-time	Fourier Transform Computation & MATLAB Implementation	Computes X(jω) of every given CT signal correctly, analytically and numerically .	Computes the spectra correctly, with minor errors in scaling, sampling rate, frequency axis	Computes the spectra for only some signals, or has noticeable errors in code, scaling	Unable to compute the Fourier transform or produce meaningful	20	2	

signals and analyze their significant spectral components.		Uses a suitable time range, sampling rate and frequency axis in rad/s or Hz.	or parameters.	or frequency axis, and needs assistance.	spectra.			
	Magnitude & Phase Spectrum Plots	Plots $\|X(j\omega)\|$ and $\angle X(j\omega)$ correctly for every signal, with proper scaling, titles, axis labels and grid. Handles phase wrapping and phase near zero-magnitude points correctly.	Plots both spectra correctly, with minor errors in scaling, labelling, phase wrapping or axis range.	Plots the spectra with noticeable errors in scaling, phase handling or labelling, and needs guidance.	Unable to plot the magnitude and phase spectra.	20	2	

	Spectral Component Analysis	**Correctly identifies dominant frequencies, peaks, bandwidth and the frequency range holding most of the signal energy. Uses symmetry (even magnitude, odd phase for real signals) and relates the spectral lines or lobes to the signal parameters.**	**Identifies most significant spectral components and bandwidth, with minor errors or limited justification.**	**Identifies only basic spectral components. Bandwidth or symmetry reasoning is incomplete, and needs guidance.**	**Unable to identify or justify the significant spectral components.**	**20**	**2**	

	Analytical vs Numerical Verification	**Compares the analytical and MATLAB spectra, quantifies the difference and explains its causes such as truncation, sampling and spectral leakage. Shows how signal parameters change the spectrum.**	**Compares analytical and numerical results correctly, with brief explanation of differences.**	**Makes a basic comparison with noticeable errors, and the explanation of differences is incomplete.**	**Unable to verify the numerical spectra against the analytical results.**	**20**	**2**	
	Engineering Analysis, Record & Interpretation	**Provides technically accurate interpretation of the**	**Maintains a proper record with correct interpretation, with**	**Record and interpretation are partial or only descriptive,**	**Incomplete record, limited or incorrect interpretation**	**20**	**2**	

		results and relates them to the physical or engineering meaning of the signal. Maintains a complete, well-organized record with observations, tables, plots and conclusions.	**minor omissions in explanation or conclusions.**	**and need guidance.**	**of the results.**			
					Total	**100**	**10**	